```
SELECT rp.perms
  FROM role_permissions rp
  JOIN user_roles ur
    ON ur.role_id = rp.role_id
 WHERE ur.space_id = ? AND ur.user_id = ?
```

Searching the database for roles and permissions can be expensive, but the current implementation will repeat this work every time the `requirePermission` filter is called, which could be several times while processing a request. To avoid this issue and simplify the logic, you can extract the permission look up into a separate filter that runs before any permission checks and stores the permissions in a request attribute. Listing 8.8 shows the new `lookupPermissions` filter that performs the mapping from user to role to permissions, and then updated `requirePermission` method. By reusing the existing permissions checks, you can add RBAC on top without having to change the access control rules. Open UserController.java in your editor and update the `requirePermission` method to match the listing.

Listing 8.8 Determining permissions based on roles

```
public void lookupPermissions(Request request, Response
response) {
    requireAuthentication(request, response);
    var spaceId = Long.parseLong(request.params(":spaceId"));
    var username = (String) request.attribute("subject");

    var perms = database.findOptional(String.class,    #A
            "SELECT rp.perms " +    #A
            "  FROM role_permissions rp JOIN user_roles ur"
+    #A
```